

10-Minute Presentation Script

TinyML-Based Classroom Occupancy Estimation for Energy-Efficient Building Applications

0:00-0:20 Slide 1 - Cover

Good morning/afternoon everyone. My name is Ahmed Mohamed Abdalla Ahmed. Today, I will present my project, TinyML-Based Classroom Occupancy Estimation for Energy-Efficient Building Applications.

The project uses a low-cost, low-voltage embedded system to estimate room occupancy locally and provide energy-aware feedback.

0:20-0:50 Slide 2 - Graphical Abstract

Before going into detail, this graphical abstract summarises the complete project workflow.

We begin with indoor sensors, collect labelled data for EMPTY, LOW, and HIGH occupancy, convert PIR motion into numerical features, train a lightweight TinyML model, and analyse it for deployment on the STM32. Finally, the occupancy result is combined with light information to provide a local energy-saving reminder.

0:50-1:25 Slide 3 - Introduction

Indoor rooms consume energy through lighting, air-conditioning, and other equipment while occupancy changes over time.

The problem is that a room may become empty or lightly occupied while energy-consuming equipment remains active. Therefore, occupancy information can help identify possible unnecessary energy use.

This project does not directly control building equipment. Instead, it provides local decision support by estimating room occupancy and giving a warning when there may be an opportunity to save energy.

1:25-2:05 Slide 4 - Problem Statement

The problem has three main parts.

First, lighting or air-conditioning may remain ON after people leave a room. Second, the ON or OFF state of equipment does not tell us whether the room is EMPTY, LOW occupancy, or HIGH occupancy. Third, the solution must work locally within the memory and processing limits of an STM32 microcontroller. It must also remain a safe low-voltage prototype without direct connection to building mains equipment.

2:05-2:40 Slide 5 - Research Questions

Based on this problem, the research questions cover the complete path from sensing to feedback.

They ask how low-cost sensors can be integrated with the STM32, how reliable labelled occupancy data can be collected, how PIR motion can distinguish occupancy conditions, how a compact AI model can fit the STM32, and how occupancy information can support energy-aware feedback safely.

2:40-3:05 Slide 6 - Project Objectives

The project has three objectives.

First, to build and validate a low-voltage sensing system and collect labelled occupancy data. Second, to develop and evaluate a lightweight TinyML occupancy classifier using PIR motion features. Third, to convert the model into INT8 format, analyse it for STM32 deployment, and connect its result to local feedback.

3:05-3:30 Slide 7 - Methodology Overview

The methodology followed one connected workflow.

I integrated the sensors, collected labelled sessions, cleaned and prepared the data, extracted motion features, trained the model, and then analysed the model for embedded deployment.

An important point is that the AI model uses PIR motion features only. The other sensors provide room monitoring and energy context.

3:30-4:05 Slide 8 - System Architecture

The system uses two PIR sensors to observe motion from different directions.

The BH1750 measures ambient light, while the BME280 measures temperature and humidity. The VL53L0X provides short-range distance context.

The STM32 reads the sensors, handles the buttons, displays information on the OLED, and controls the LEDs and buzzer. The complete prototype operates at low voltage and does not connect directly to building mains power.

4:05-4:25 Slide 9 - Portable Occupancy Monitoring Prototype

This slide shows the physical prototype.

It integrates the STM32 board, sensors, buttons, OLED display, LEDs, and buzzer in one portable enclosure. I used this prototype for sensor testing, labelled data collection, and the live demonstration.

4:25-5:00 Slide 10 - Data Collection and Labelling

During each recording session, the STM32 sampled the sensors approximately once per second and sent labelled records through UART to a CSV logger.

Button 1 selected the occupancy label: EMPTY for no person, LOW for one person, and HIGH for two or three people. Button 2 started and stopped each session. I also recorded sessions under different lighting and air-conditioning conditions to avoid training the model for only one room condition.

5:00-5:35 Data Preparation and PIR Features

The raw data was not sent directly to the model.

I checked the labels, removed transition samples at the beginning and end of sessions, and retained valid records only.

Then, I built motion windows from the latest 30 samples, which represent approximately 30 seconds. From each window, I calculated six PIR features: activity ratios for PIR 1 and PIR 2, any-PIR activity, both-PIR activity, and rising-edge counts for each PIR.

This allows the model to interpret a motion pattern over time instead of reacting to one PIR reading.

5:35-6:15 TinyML Training

The neural network has six inputs, followed by hidden layers with 16 and 8 ReLU neurons, and three outputs: EMPTY, LOW, and HIGH.

I separated whole recordings across training, validation, and test sets. Therefore, the test sessions were not previously seen by the model. I also used early stopping so that training stopped when validation performance no longer improved.

6:15-7:25 Results and Discussion

The final model was tested using 256 motion windows from unseen recording sessions.

It achieved 88.67% test accuracy, meaning that it correctly classified 227 out of 256 test windows. The macro F1 score was 87.47%. This confirms that the performance was reasonably balanced across EMPTY, LOW, and HIGH occupancy classes.

The model classified all 102 EMPTY windows correctly and all 57 LOW windows correctly. For HIGH occupancy, it correctly classified 68 out of 97 windows. Most errors occurred when HIGH occupancy was predicted as LOW. This indicates that the next dataset should include more varied HIGH-occupancy motion recordings.

The INT8 TensorFlow Lite model file is only 3,616 bytes. STM32Cube.AI analysis reported 320 MACC per inference, 13.60 KiB Flash, and 2.91 KiB RAM. These requirements are small compared with the resources available on the NUCLEO-L476RG.

7:25-8:05 Engineering Decisions and Limitations

The engineering decisions were deliberate.

Two PIR sensors were used because motion provides the most direct evidence for occupancy. BH1750 and BME280 readings support energy-aware rules, while VL53L0X remains distance context because its coverage is too narrow to represent the entire room.

The main limitation is that PIR sensors detect movement, not people directly. A person who remains very still may appear similar to an EMPTY condition. Also, HIGH occupancy has more varied movement patterns, which explains why some HIGH windows were classified as LOW.

8:05-8:40 Live Demonstration

I will now demonstrate the prototype.

First, I will power on the STM32 system and confirm that the sensors are ready. Next, I will create motion near the PIR sensors and observe the occupancy status shown by the prototype. Finally, when EMPTY occupancy is detected while ambient light remains high for the required time, the system provides a local warning through the OLED, LEDs, and buzzer.

8:40-9:25 Conclusion and Future Work

In conclusion, this project demonstrated that a low-cost, low-voltage STM32 system can collect motion data, estimate classroom occupancy locally, and provide energy-aware feedback.

The 88.67% test accuracy provides a validated baseline for further development.

Future work includes adding a third PIR sensor to improve coverage, collecting more varied HIGH-occupancy data, evaluating sensor fusion with sufficient new data, and improving the prototype enclosure.

9:25-9:40 References

These references include the STM32 board documentation, STM32Cube.AI documentation, TensorFlow Lite resources, and the datasheets for the sensors used in this project.

9:40-10:00 Closing

Thank you for your attention. I am ready for your questions.